

DecodeLabs Internship — Project 1

Data Cleaning Change Log

Purpose: Document what was changed in the dataset, why the change was made, and the validation performed after cleaning.

Change ID	Description	Impact	Status
CL001	Replaced 309 missing values in CouponCode with Unknown .	Removed all missing values from CouponCode while preserving all 1,200 records.	Resolved
VAL001	Validated the Date column and confirmed consistent datetime formatting.	Date column retained as datetime64[ns] ; no correction required.	Verified
VAL002	Validated TotalPrice against $\text{Quantity} \times \text{UnitPrice}$.	Initial apparent differences were resolved by rounding to two decimal places; all records matched and no values were changed.	Verified
VAL003	Checked categorical columns for unexpected values, capitalization issues, spelling variations, and leading/trailing spaces.	No inconsistencies found in Product, PaymentMethod, OrderStatus, CouponCode, or ReferralSource.	Verified
VAL004	Checked duplicate rows and identifier uniqueness.	0 duplicate rows, 0 duplicate OrderIDs, and 0 duplicate TrackingNumbers.	Verified
VAL005	Checked identifier completeness.	No missing OrderID, CustomerID, or TrackingNumber values.	Verified

Final validation: 1,200 records $\times$ 14 columns; 0 missing values; 0 duplicate rows; 0 duplicate OrderIDs; 0 duplicate TrackingNumbers. The cleaned dataset was exported as **Cleaned_Dataset.xlsx**.